Trong-Tung Nguyen

Ph.D. Student in Computer Science · Data Science & AI Institute
Whiting School of Engineering
Johns Hopkins University

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Research

My past research focused on efficient generative models for visual generation: image editing, inpainting, and personalization. My current research explores physics-informed video generation models, which enable physics realism in generated videos, and 3D physics simulation, which enables dynamic physics on static 3D representations.

Education

Johns Hopkins University, Whiting School of Engineering, USA
Ph.D. Student in Computer Science — 2026 – present
Advisor: Anand Bhattad
Research Topics: Physics-informed Video Generation, Physics-informed 3D Generation & Efficient Visual Generation Model

Ho Chi Minh University of Science, Vietnam National University, Ho Chi Minh City, Vietnam
B.Sc., Computer Science — Advanced Program in Computer Science (APCS) — 2017 – 2021
GPA: 3.92/4.0, Math 4.0/4.0 — **Top 1%** among graduated students

Research Experience

Qualcomm AI Research (formerly VinAI Research), AI Research Resident — 2023 – 2026
Mentors: Dr. Anh Tran, and Prof. Cuong Pham

Singapore Management University, Visiting Researcher — Jan 2023 – Aug 2023
Mentor: Prof. Guansong Pang

VinUni–Illinois Smart Health Center, Research and Teaching Assistant — 2021 – 2022
Mentors: Prof. Minh Do, and Prof. Wray Buntine

Selected Honors and Awards

Outstanding Reviewer, **Top 5% of Reviewers**, CVPR — 2026
Best AI Resident Award, Qualcomm AI Research Residency Program — 2025
Travel Support Award, CVPR — 2025
Best Presentation Award, The 10th International Symposium on Information and Communication Technology (SoICT) — 2019
Finalist, National Codewar Competition, top 5, hosted by FSoft — 2019
Global Software Talent Scholarship, FPT Software — 2019
Runner-up, UNESCO Hackathon, Fossasia — 2018
Silver Medal, 30/4 Southern Vietnam Mathematics Olympiad — 2016
Silver Medal, 30/4 Southern Vietnam Mathematics Olympiad — 2015
Third Prize, Mathematics Competition for Gifted High School Students, Ho Chi Minh City — 2014

Publications and Selected Preprints

* *denotes equal contribution*

9. **PowerSim: Differentiable Physics Simulation and Rendering with Power Diagrams**
   Trong-Tung Nguyen, Anand Bhattad
   ON ARXIV 2026
8. **FracGen: Learning How Objects Stretch and Tear with Physics-Informed Video Generation**
   Trong-Tung Nguyen, Jiahan Zhang, Anand Bhattad
   ON ARXIV 2026
7. **InverFill: One-Step Inversion for Enhanced Few-Step Diffusion Inpainting**
   Duc Vu*, Kien Nguyen*, **Trong-Tung Nguyen***, Ngan Nguyen*, Phong Nguyen, Khoi Nguyen, Cuong Pham, Anh Tran
   CVPR 2026
6. **SwiftPie: Lightning-fast Subject-driven Image Personalization via One step Diffusion**
   Huy Duong*, **Trong-Tung Nguyen***, Cuong Pham, Anh Tran, Khoi Nguyen, Minh Hoai
   CVPR 2026 (FINDINGS)
5. **SwiftEdit: Lightning Fast Text-guided Image Editing via One-step Diffusion**
   Trong-Tung Nguyen, Quang Nguyen, Khoi Nguyen, Anh Tran, Cuong Pham
   CVPR 2025

4. **FlexEdit: Flexible and Controllable Diffusion-based Object-centric Image Editing**
   Trong-Tung Nguyen, Duc-Anh Nguyen, Anh Tran, Cuong Pham
   ON ARXIV 2024

3. **Multi-stream Fusion for Class Incremental Learning in Pill Image Classification**
   Trong-Tung Nguyen, Hieu H. Pham, Phi Le Nguyen, Thanh Hung Nguyen, Minh Do
   ACCV 2022

2. **iTASK — Intelligent Traffic Analysis Software Kit**
   Minh-Triet Tran, Tam V. Nguyen, Trung-Hieu Hoang, Trung-Nghia Le, Khac-Tuan Nguyen, Dat-Thanh Dinh, Thanh-An Nguyen, Hai-Dang Nguyen, Xuan-Nhat Hoang, **Trong-Tung Nguyen**, Viet-Khoa Vo-Ho, Trong-Le Do, Lam Nguyen, Minh-Quan Le, Hoang-Phuc Nguyen-Dinh, Trong-Thang Pham, Xuan-Vy Nguyen, E-Ro Nguyen, Quoc-Cuong Tran, Hung Tran, Hieu Dao, Mai-Khiem Tran, Quang-Thuc Nguyen, Tien-Phat Nguyen, The-Anh Vu-Le, Gia-Han Diep, Minh Do
   CVPR 2020 WORKSHOPS

1. **Visual Assistant for Crowdsourced Anomaly Event Recognition in Smart City**
   Minh-Tri Ho, **Trong-Tung Nguyen**, Hieu Dao, Minh-Triet Tran
   SoICT 2019

Academic Services

Reviewer

- CVPR (25, 26), ECCV (24, 26), NeurIPS (26), ICLR (27), AAAI (25, 26)

Teaching Assistant

- Intro to AI (VinUni — Spring 2022), Learning-based Vision (Johns Hopkins University — Spring 2026)